A Multifunctional 28-nm 4-Kb 9T1C SRAM-CIM Macro With Bidirectional Read/Write, Full-Array Self-Cycling Boolean Logic, and MAC Operation

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***Abstract*—This article presents a 28-nm 4-Kb 9T1C SRAM-CIM macro that unifies bidirectional read/write, full-array self-cycling Boolean logic operations, and charge-domain multiply–accumulate (MAC) operation within a compact bit-cell structure. The proposed cell supports both row- and column-wise access, where the row-wise read path effectively suppresses read disturbance and improves read stability. In addition, AND, OR, and IMP operations are executed directly in memory through a self-cycling mechanism, while MAC operation is realized by charge-domain accumulation using the embedded capacitor. Simulation results show that the proposed cell maintains stable data-holding capability under different temperatures and process corners, and the row-wise RSNM is significantly higher than that of column-wise read and conventional 6T read. The proposed Boolean logic operations exhibit stable energy behavior under process and voltage variations, and 2000 Monte Carlo simulations verify 100% correctness for representative cases. The MAC outputs also show near-linear behavior, with the absolute INL within 0.026–0.08 LSB, and stable output distributions under process variation. The proposed macro achieves logic energy as low as 1.218 fJ/bit, Boolean logic energy efficiency of 821 TOPS/W, and Boolean logic throughput of 1024 GOPS/Kb, while the MAC throughput reaches 1170.3 GOPS with an energy efficiency of 696–1323.5 TOPS/W. These results indicate that the proposed design provides a compact and efficient multifunctional SRAM-CIM solution for both logic-centric and MAC-oriented applications.**

***Index Terms*—Bidirectional read/write, computing-in-memory (CIM), multiply–accumulate (MAC), self-cycling Boolean logic, static random access memory (SRAM).**

I. Introduction

In RECENT years, the rapid development of artificial intelligence (AI) and data-driven applications has led to a dramatic increase in the demand for large-scale data processing. Modern workloads frequently require repeated data exchange between the memory and processing units, which becomes increasingly costly in conventional von Neumann architectures where storage and computation are physically separated. Such frequent data movement causes the well-known memory-wall bottleneck, resulting in substantial penalties in both latency and energy consumption [1], [2]. This challenge has motivated the development of computing-in-memory (CIM), which performs part of the computation directly inside memory arrays to reduce data transfer overhead. Among the available memory technologies, static random access memory (SRAM) is particularly attractive for CIM because of its high access speed, low power consumption, strong reliability, and good scalability in modern computing systems [3], [4], [5].

From an application perspective, SRAM-CIM has mainly evolved along two major directions. One is neural-network acceleration, where convolutional neural networks (CNNs) require a massive number of multiply–accumulate (MAC) operations and therefore benefit from in-memory arithmetic execution [6], [7], [8], [9]. The other is logic-centric acceleration, represented by the Advanced Encryption Standard (AES), which relies heavily on bitwise Boolean operations, especially repeated XOR-related logic across multiple encryption rounds [10].

Significant progress has been achieved in both directions. For MAC-oriented CIM, many prior works improve charge-domain accumulation efficiency, weighted-summation parallelism, and precision controllability, thereby enabling efficient neural-network inference inside SRAM arrays [11], [12], [13], [14]. For logic-oriented CIM, digital-domain SRAM architectures provide strong computational robustness and accuracy [15], [16], [17], while self-cycling Boolean logic operations and in-array data manipulation mechanisms have been developed to accelerate cryptographic and logic-intensive workloads [18], [19]. In parallel, bidirectional SRAM arrays and transposable memory structures have also been investigated to support more flexible data movement and multidirectional data access within memory arrays, thereby facilitating matrix transposition and other data-reorganization operations [20], [21], [22], [23].

With the continued development of SRAM-CIM, recent designs have increasingly focused on integrating more functions within a single memory-computing framework, rather than optimizing only one specific operation. For example, Peng et al. [11] propose a dual-mode SRAM-CIM structure that can be used for CNN-based image recognition and grayscale edge detection. Lin et al. [18] realize bidirectional

Received 14 April 2026; revised 1 July 2026 and 22 July 2026; accepted 22 July 2026. Date of publication 30 July 2026; date of current version 29 September 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62471003 and Grant 92464203 and in part by the University Synergy Innovation Program of Anhui Province under Grant GXXT-2023-003 and Grant GXXT-2023-011. *(Corresponding author: Wenjuan Lu.)*

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Digital Object Identifier 10.1109/TVLSI.2026.3717001

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read/write, self-cycling Boolean logic operations, and in-array data copy in a bidirectional SRAM array. In addition, [24] integrates a CNN-oriented computing mode with an analysis mode that supports multibit multiplication and in-array result storage. Although these studies have broadened the functional scope of SRAM-CIM, the supported function sets are still not sufficiently comprehensive. Existing works usually cover only part of the desired function set, while a unified SRAM-CIM design that simultaneously supports MAC operation for CNN acceleration, bidirectional read/write, and self-cycling Boolean logic operations is still lacking. These functionalities impose conflicting requirements on storage-node stability, charge-domain accumulation behavior, self-cycling result restoration, and access topology.

To address these challenges, this work proposes a 28-nm 9T1C SRAM-based CIM macro that, through reconfigurable operating modes, supports row- and column-wise bidirectional read/write, full-array self-cycling Boolean logic operations, and charge-domain MAC operation within the same compact 9T1C structure.

The main contributions of this work are summarized as follows.

1) *A Bidirectional 9T1C SRAM Cell:* The proposed cell supports row- and column-wise read/write. In particular, the row-wise read path provides improved noise margin and avoids read disturbance.

2) *High-Efficiency Self-Cycling Boolean Logic Operations:* The proposed macro supports full-array self-cycling Boolean logic operations, including AND, OR, and IMP, with high throughput and energy efficiency.

3) *Compact Multifunctional in-Memory Computing:* By reusing transistors within a compact cell structure, the proposed design supports bidirectional data remapping, self-cycling Boolean logic operations, and MAC operation in the same SRAM-CIM macro.

The remainder of this article is organized as follows. Section II presents the proposed SRAM-CIM architecture and operating principles. Section III reports simulation results and performance evaluations under different operation modes. Section IV concludes this article.

II. Proposed SRAM-CIM Macro

A. Overall Architecture of the Proposed SRAM-CIM Macro

As shown in Fig. 1(a), the proposed SRAM-CIM macro mainly consists of a 9T1C memory array together with peripheral modules, including MAC and logic control circuits, Boolean logic driver circuits, read/write circuits, sense-amplifier (SA) and precharge circuits, DAC and MAC driver circuits, SAR ADCs and time control circuits. The 9T1C memory array contains 64 columns, where each column is vertically composed of four block units. Each block unit consists of 16 cells, resulting in a total array capacity of 4 Kb.

The operation modes of the CIM macro, including bidirectional read/write, self-cycling Boolean logic operations, and MAC operation, are coordinated by the peripheral control circuits, which generate the corresponding control signals for the memory array and peripheral modules.

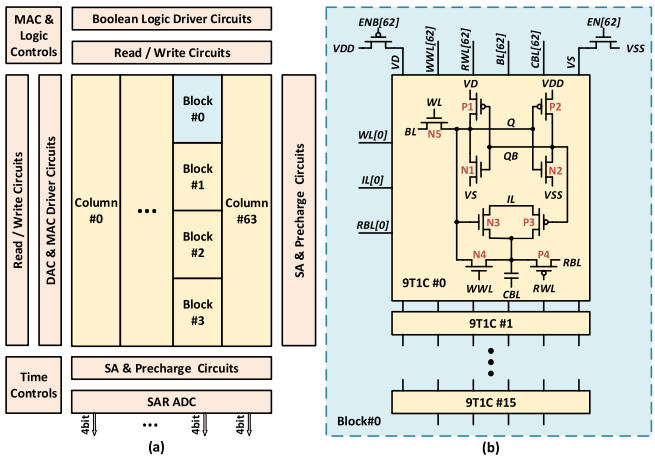


Fig. 1. (a) Overall architecture of the proposed SRAM-CIM macro. (b) Block-level organization and 9T1C SRAM cell structure.

As illustrated in Fig. 1(b), each block unit consists of 16 9T1C cells connected along the column direction. These 16 cells share MOS transistors controlled by the column-wise signals EN and ENB, which selectively connect nodes VD and VS to VDD and VSS, respectively. This shared structure enables efficient column-wise control of multiple cells, thereby supporting efficient single-ended sensing and fast read/write operations while also improving the execution efficiency of self-cycling Boolean logic operations.

The proposed 9T1C cell employs a pair of cross-coupled inverters composed of transistors P1, P2, N1, and N2, which form the bistable storage nodes Q and QB. Additional access transistors together with a MOM capacitor are introduced to support multimode computing operations. The cell provides multiple data access paths through BL, RBL, and IL, enabling flexible data transfer for both row-wise and column-wise operations. The column-wise control signals EN and ENB are used to temporarily disconnect the storage nodes from the power rails during specific operation modes, allowing the bistable latch to be reconfigured for bidirectional read/write operations or self-cycling Boolean logic operations.

By reusing the same access devices across different operation modes, the proposed structure supports bidirectional read/write, self-cycling Boolean logic operations, and MAC operation within the same compact cell architecture. The detailed operating principles of these functions are described in Sections II-B–II-D.

B. Bidirectional Read and Write Operations

The proposed 9T1C cell supports column-wise read/write operations similar to those of a conventional 6T SRAM, while an additional row-wise access path is introduced to enable bidirectional data access within the memory array.

During the column-wise write operation, as illustrated in Figs. 2(a) and 3(a), the connections between IL/RBL and the cell are first disabled through WWL. Meanwhile, EN is driven low, and ENB is driven high, thereby disconnecting nodes VD and VS from VDD and VSS, respectively. As a result, the original bistable storage structure is temporarily converted into

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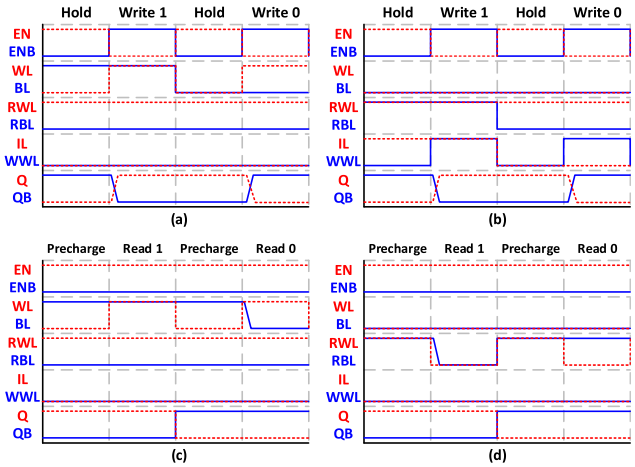


Fig. 2. Timing diagrams illustrating different access operations of the proposed 9T1C SRAM cell. (a) Column-wise write operation. (b) Row-wise write operation. (c) Column-wise read operation. (d) Row-wise read operation.

an inverter. Subsequently, WL is driven to VDD, turning on the transistor N5, and the write data are applied to the inverter input through BL. After the write operation is completed, EN and ENB are re-enabled, reconnecting VD to VDD and VS to VSS, thereby restoring the bistable storage structure and completing the write process.

Similarly, during the row-wise write operation, as shown in Figs. 2(b) and 3(b), the connection between BL and the cell is disabled by WL. The write operation is then performed under the control of WWL after the bistable storage structure is temporarily broken. The conduction of IL depends on the stored logic states of Q and QB. When Q = 1, the write data are delivered through both IL and RBL; when Q = 0, the write data are delivered only through RBL. The data are then applied to the internal storage node through transistor N4 controlled by WWL, thereby completing the write operation.

During the column-wise read operation, as illustrated in Figs. 2(c) and 3(c), BL is precharged to VDD. Subsequently, WL is driven to VDD, allowing the stored value at node Q to be transferred to BL. BL remains connected to the OUT terminal of the column-wise SA located at the bottom of the array, while the BOUT terminal is biased at 800 mV. After a certain sensing interval, WL is driven to VSS, thereby disconnecting node Q from BL. The resulting voltage difference is then amplified by the column-wise SA, thereby completing the read operation.

During the row-wise read operation, as shown in Figs. 2(d) and 3(d), RBL is precharged to VDD, while IL is tied to VSS. During the read phase, RWL is driven to VSS. When Q = 1, RBL is discharged through IL; when Q = 0, RBL remains at the precharged high level. RBL remains connected to the BOUT terminal of the row-wise SA located on the right side of the array, while the OUT terminal is biased at 800 mV. Under this configuration, when Q = 1, the BOUT voltage becomes lower than OUT, resulting in a logic "1" at the output; when Q = 0, the BOUT voltage is higher than OUT, resulting in a logic "0." The remaining readout procedure is similar to that

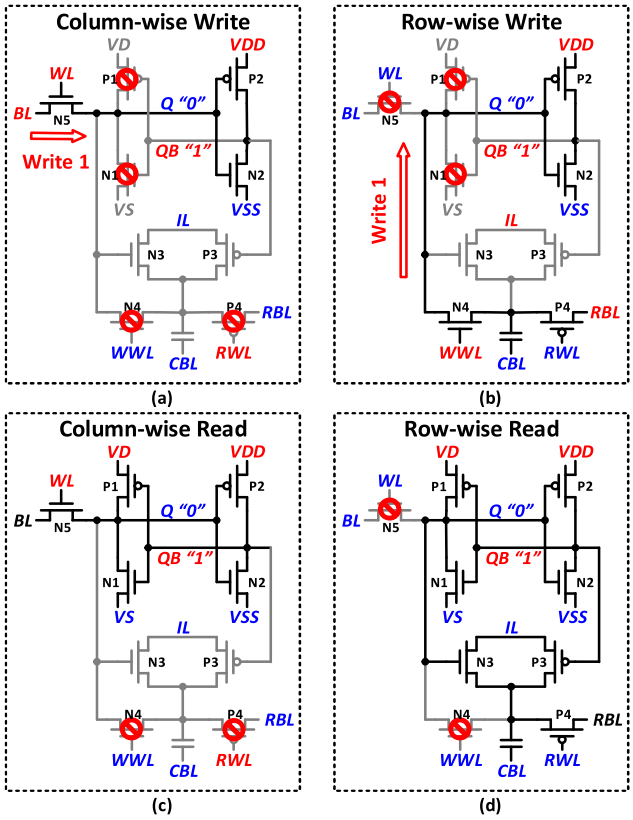


Fig. 3. Illustration of the bidirectional read and write operations in the proposed 9T1C SRAM-CIM cell. (a) Column-wise write operation. (b) Row-wise write operation. (c) Column-wise read operation. (d) Row-wise read operation.

of the column-wise read operation and is therefore omitted for brevity.

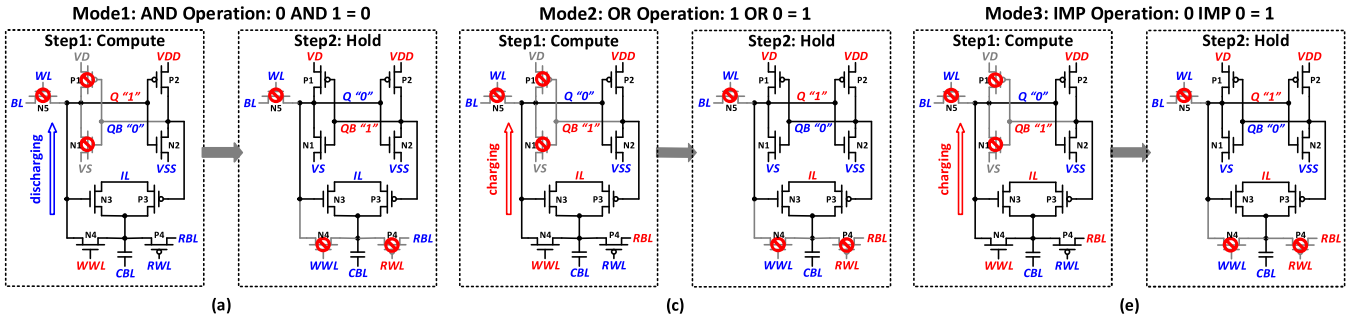
It is worth noting that during the row-wise read operation, WWL is always driven to VSS, keeping transistor N4 turned off. As a result, the discharge path from RBL to IL does not introduce voltage disturbance at the storage nodes. This design prevents read disturb while improving the read noise margin.

Moreover, the bidirectional row- and column-wise access paths also enable in-array matrix transposition and data reorganization. Since data can be read along one direction and written back through the orthogonal direction, the proposed macro can efficiently transform data mappings between rows and columns without requiring dedicated transpose circuitry or additional data-movement hardware. Although transistor reuse introduces additional energy overhead due to the charging and discharging of the embedded MOM capacitor, this capability further improves the flexibility of the proposed SRAM-CIM macro for dataflow adaptation in different in-memory computing tasks.

C. Self-Cycling Boolean Logic Operation

In addition, the proposed 9T1C cell can perform in-memory Boolean logic operations through a unified operation mechanism. By temporarily breaking the bistable storage structure and reusing the existing access paths, logic computation can be executed directly within the memory array without requiring additional logic circuitry. After the computation phase, the

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Mode1: AND Operation: Truth Table

Operand 1	Operand 2	WWL	Q(n)	Q(n+1)
0	0	VDD	0	0
0	1	VDD	1	0
1	0	VSS	0	0
1	1	VSS	1	1

(b)

Mode2: OR Operation: Truth Table

Operand 1	Operand 2	WWL	Q(n)	Q(n+1)
0	0	VSS	0	0
0	1	VSS	1	1
1	0	VDD	0	1
1	1	VDD	1	1

(d)

Mode3: IMP Operation: Truth Table

Operand 1	Operand 2	WWL	Q(n)	Q(n+1)
0	0	VDD	0	1
0	1	VDD	1	1
1	0	VSS	0	0
1	1	VSS	1	1

(f)

Fig. 4. Illustration of the self-cycling Boolean logic operations in the proposed 9T1C SRAM-CIM cell. (a) Operation procedure of AND. (b) Truth table of AND. (c) Operation procedure of OR. (d) Truth table of OR. (e) Operation procedure of IMP. (f) Truth table of IMP.

logic result can be restored and latched back into the cell through the self-cycling process. Fig. 4 illustrates the operation procedures and truth tables of the supported Boolean operations, including AND, OR, and IMP.

The full-array self-cycling operation is achieved by applying the same operation timing to the shared control lines of the array. During a Boolean logic cycle, all selected cells enter the computation phase simultaneously by disconnecting VD and VS from VDD and VSS through EN and ENB. The stored data in each cell serves as one operand, while the input operand is applied through the corresponding control condition of WWL, IL, and RBL. After the local logic result is generated at node Q, EN and ENB restore the cross-coupled inverter structure, and the computed result is regenerated and latched back into the same cell. Therefore, no intermediate readout or external write-back path is required during the logic operation, and the same compute-and-restore procedure can be executed in parallel across the full array.

As shown in Fig. 4(a), the AND operation consists of two phases: the computation phase and the restoration phase. During the computation phase, EN is driven low and ENB is driven high, thereby disconnecting nodes VD and VS from VDD and VSS and temporarily breaking the bistable structure formed by the cross-coupled inverters (N1–N2 and P1–P2). Meanwhile, IL and RBL are driven to VSS, and RWL is asserted low to maintain the connection between RBL and the upper plate of the storage capacitor. WL remains at VSS to turn off N5 and isolate node Q from BL.

Under this configuration, the Boolean computation is determined by the interaction between the stored operand and the input operand. When the input operand is logic "0," WWL is driven to VDD, allowing the low voltage stored on the capacitor to be written to node Q. When the input operand is logic "1," WWL remains at VSS, and node Q retains its value through the gate capacitances of N2 and P2. In this way, the AND truth table shown in Fig. 4(b) is realized.

After the computation phase, EN is driven high and ENB is driven low, reconnecting VD to VDD and VS to VSS. The cross-coupled inverters then recover their bistable structure, and the computed result is latched at nodes Q and QB. In addition, the conduction states of N3 and P3 depend on the stored data at nodes Q and QB, which determines the effective charging path of the storage capacitor. When Q = 1, both IL and RBL contribute to charging the capacitor, whereas when Q = 0, the computation path is provided only through RBL.

The OR and IMP operations follow the same self-cycling procedure, with only the WWL driving conditions adjusted according to their respective truth tables. For both operations, RBL and IL are driven to VDD. In the OR operation shown in Fig. 4(c) and (d), WWL remains at VSS when the input operand is logic "0," preserving the stored value, while WWL is driven to VDD when the input operand is logic "1," writing the capacitor voltage to node Q. In contrast, for the IMP operation shown in Fig. 4(e) and (f), WWL is driven to VDD when the input operand is logic "0," whereas it remains at VSS when the input operand is logic "1," thereby preserving the stored data.

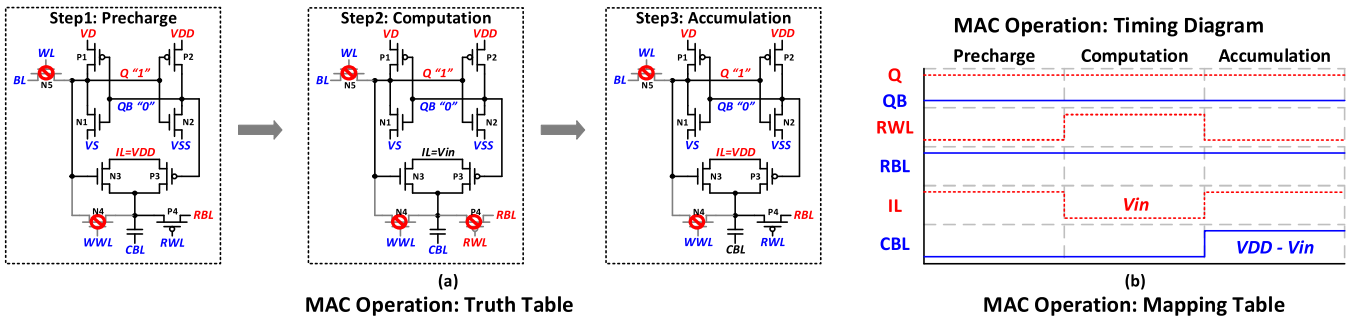
It is worth noting that, since the array supports both AND and OR operations, XOR and XNOR computations can be realized within two operation cycles.

Therefore, by reusing the existing access transistors and storage capacitor, the proposed 9T1C SRAM-CIM cell enables efficient in-memory execution of multiple Boolean logic operations through a unified self-cycling mechanism while maintaining full SRAM read/write functionality without requiring additional logic hardware.

D. MAC Operation

Fig. 5(a) illustrates the operation procedure of the MAC computation. During the entire MAC process, EN remains high and ENB remains low, maintaining the connection of VD to VDD and VS to VSS so that node Q keeps its stable state. Meanwhile, WL and WWL remain low, turning off transistors N4 and N5. In addition, RBL is always driven to VDD.

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MAC Operation: Truth Table

	Q = 0 / QB = 0.9			Q = 0.9 / QB = 0		
Process	Precharge	Computation	Accumulation	Precharge	Computation	Accumulation
RWL	VSS	VDD	VSS	VSS	VDD	VSS
RBL	VDD	VDD	VDD	VDD	VDD	VDD
IL	VDD	Vin	VDD	VDD	Vin	VDD
CBL	VSS	VSS	VSS	VSS	VSS	Vout

(c)

MAC Operation: Mapping Table

IN	Vin (mV)	IN	Vin (mV)
0	900	8	420
1	840	9	360
2	780	10	300
3	720	11	240
4	660	12	180
5	600	13	120
6	540	14	60
7	480	15	0

(d)

Fig. 5. Illustration of the MAC operation in the proposed SRAM-CIM architecture. (a) MAC operation procedure. (b) Timing diagram of the MAC operation. (c) Truth table of the MAC operations. (d) Mapping table for MAC operations.

As shown in Fig. 5(b) and (c), the MAC operation consists of three phases: precharge, computation, and accumulation, which are described as follows.

During the precharge phase, IL and RBL are driven to VDD, while RWL is driven low, enabling RBL to charge the upper plate of the storage capacitor through the corresponding path. In addition, CBL is driven to VSS during this phase.

During the computation phase, RWL is driven high, turning off P4 and thereby disconnecting the charging path from RBL to the capacitor. IL is driven by the DAC to the analog input voltage V_{in}, which corresponds to the 4-bit input value of the MAC operation, as shown in Fig. 5(d). The stored value at node Q represents the 1-bit weight and determines whether the input voltage can be transferred to the upper plate of the capacitor. When Q = 1, transistors N3 and P3 are turned on, allowing the capacitor voltage to follow V_{in}. When Q = 0, N3 and P3 remain off, and the capacitor voltage stays at the precharged level VDD.

During the accumulation phase, the VSS driving of CBL is released. At the same time, IL is disconnected from the DAC and driven to VDD, while RWL is driven low to turn on P4, allowing the upper plate of the capacitor to be charged back to VDD. Since all cells perform the above operations simultaneously, the voltage accumulated on CBL follows the relationship described in (1), which represents the MAC result of 64 four-bit inputs with one-bit weights. The accumulated result is then converted to digital form by a 4-bit SAR ADC for readout

$$V_{CBL} = \frac{\sum_{i=0}^{63}(VDD - Vx_i)C}{64C}\,(Vx_i = VSS\,or\,V\,in). \quad (1)$$

Fig. 6 illustrates how 4-bit inputs and 4-bit weights in CNNs are mapped to the proposed macro to perform MAC operations. Each 4-bit weight is decomposed into four binary bits and stored in four consecutive 9T1C cells along the row direction, forming a spatial bit-plane mapping across adjacent

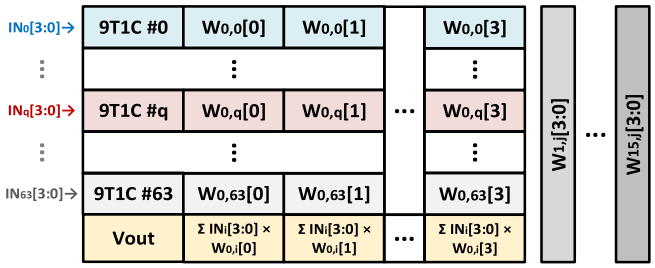


Fig. 6. Mapping of 4-bit inputs and 4-bit weights to the proposed macro for MAC operation in CNNs.

columns. This arrangement allows the corresponding 1-bit-weight MAC results to be generated in parallel within one MAC cycle, rather than through multicycle bit-serial operation. In the proposed macro, each column contains up to 64 weight cells participating in the accumulation, corresponding to the accumulation of 64 partial products. Meanwhile, along the row direction, the architecture enables 16 4-bit MAC operations to be executed in parallel, thereby supporting efficient vector–matrix multiplication. The input values are converted into analog voltages by a charge-scaling DAC and applied to the IL line to participate in the MAC computation.

III. Simulation Results and Analysis

The proposed 4Kb 9T1C SRAM-CIM macro was implemented in a 28-nm CMOS process. Fig. 7(a) shows the layout of the proposed 9T1C cell, and Fig. 7(b) shows the layout of the complete 64×64 macro. The 9T1C SRAM array is placed at the center, while the DACs, SAs, and 4-bit SAR ADCs are arranged around the array to support different operation modes, with the corresponding implemented peripheral overheads reflected in the following simulation results. The embedded MOM capacitor introduces only about 3% area overhead while enabling the extended CIM functionality.

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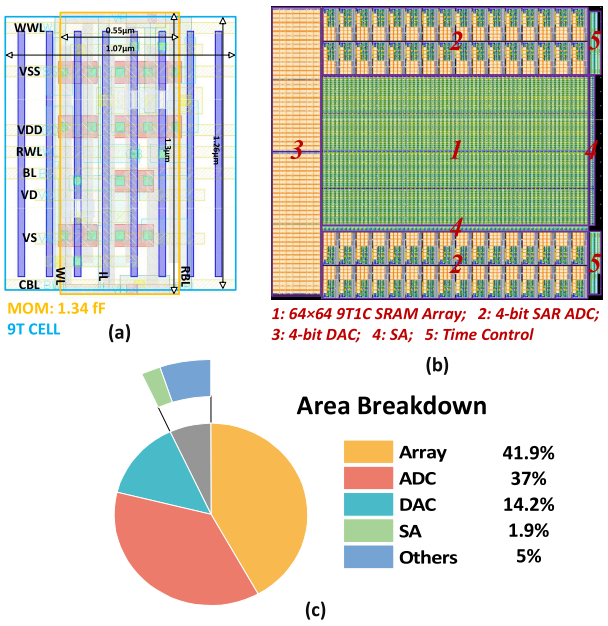


Fig. 7. Layout of the proposed 9T1C SRAM-CIM macro. (a) 9T1C cell layout. (b) Macro layout. (c) Area breakdown.

The area breakdown is shown in Fig. 7(c). The array occupies 41.9% of the total area, while the ADCs and DACs account for 37% and 14.2%, respectively. The remaining area is occupied by the SAs and other auxiliary circuits. These results indicate that the proposed macro can integrate bidirectional read/write, self-cycling Boolean logic operations, and MAC operation within a compact layout. The relatively large peripheral ratio mainly comes from the column-parallel 4-bit SAR ADCs, which are adopted to maintain 64-channel MAC readout parallelism and high throughput; ADC sharing could reduce the peripheral area but would increase circuit complexity and readout latency. Therefore, one ADC per column is adopted to maximize MAC throughput.

The stability of the proposed 9T1C cell was first evaluated by HSNM and RSNM. As shown in Fig. 8(a), the HSNM was simulated under five process corners and three temperatures. At 27 °C, the HSNM ranges from 0.3062 to 0.3295 V, while under all tested conditions it remains within 0.2747–0.3448 V, indicating stable data-holding capability of the proposed cell. As temperature increases, the HSNM decreases at all process corners, which is consistent with the expected degradation of cell stability at elevated temperature.

Fig. 8(b) compares the RSNM of the two read modes of the proposed 9T1C cell with that of a conventional 6 T cell. It can be observed that the row-wise read path consistently provides a much larger RSNM than the other two cases. For example, at the TT corner, the RSNM of row-wise read is 0.3191 V, whereas the corresponding values of column-wise read and 6 T read are 0.0971 and 0.0951 V, respectively. Across all tested corners, the RSNM of row-wise read remains within 0.3037–0.3287 V, while the column-wise read path shows RSNM values of 0.0791–0.1129 V, which are comparable to those of the 6 T cell. This result confirms that the proposed row-wise read path significantly improves read stability without sacrificing conventional column-wise read performance.

The write robustness was evaluated using the bidirectional write voltage margins of the row-wise and column-wise write operations under different process corners, as shown in Fig. 8(c). Both 0-write-1 and 1-write-0 transitions were considered. Positive voltage margins are maintained in all tested cases, with a minimum row-wise write margin of 206 mV and a minimum column-wise write margin of 358 mV. These results indicate reliable bidirectional write capability under process variations.

To further verify the read robustness, 2000 Monte Carlo simulations of the Q-node read disturbance during reading logic "0" were performed, as shown in Fig. 9(a)–(c). In the row-wise read case, the Q-node voltage remains near 0 V, with a mean value of 10.448 μV and a standard deviation of 7.721 μV. By contrast, the mean disturbed voltage increases to 200.748 mV for column-wise read and 200.333 mV for conventional 6 T read. These results confirm that the proposed row-wise read mechanism effectively suppresses read disturbance and improves RSNM.

Fig. 10 evaluates the energy consumption and correctness of the self-cycling Boolean logic operations. As shown in Fig. 10(a) and (b), the energy consumption of AND remains much lower than that of OR and IMP under different process corners and VDDs. At 0.9 V and 27 °C, the AND energy ranges from 0.163 to 0.179 fJ/bit under different process corners, while OR and IMP range from 0.826 to 1.599 fJ/bit. As VDD decreases from 0.9 to 0.7 V, the energy consumption of all three operations decreases monotonically, with AND reduced from 0.168 to 0.102 fJ/bit and OR/IMP reduced from 1.218 to 0.869 fJ/bit. These results indicate stable energy behavior under different operating conditions.

To evaluate the retention limitation during the temporary nonstable interval, the cell storage state under the worst-case leakage condition is simulated, as shown in Fig. 10(c). With Q = 1 initially stored, the cell remains in the nonstable state for 1 ns before restoring the bistable structure. The result shows that the stored data can be correctly recovered, confirming the leakage robustness of the proposed self-cycling operation.

Taking the 2000 Monte Carlo results of 0 AND 1 as an example, as shown in Fig. 10(d), the output voltage exhibits a mean value of 7.374 μV with a standard deviation of 5.377 μV, corresponding to the correct logic "0" result. In addition to the three representative cases shown in Fig. 10(d)–(f), 2000 Monte Carlo simulations were also performed for all Boolean input combinations of AND, OR, and IMP operations under the typical operating condition of TT, 0.9 V, and 27 °C, and all tested cases achieved 100% correctness. These results verify the robustness and correctness of the proposed Boolean logic operations under this operating condition.

Fig. 11 evaluates the energy efficiency, linearity, and robustness of the proposed MAC operation. As shown in Fig. 11(a), at 0.9 V and 27 °C, the reported results correspond to the overall energy consumption per bit of the MAC operation, including the contributions of the MAC array and the associated peripheral circuits such as the DAC and ADC. Under different process corners, the energy per

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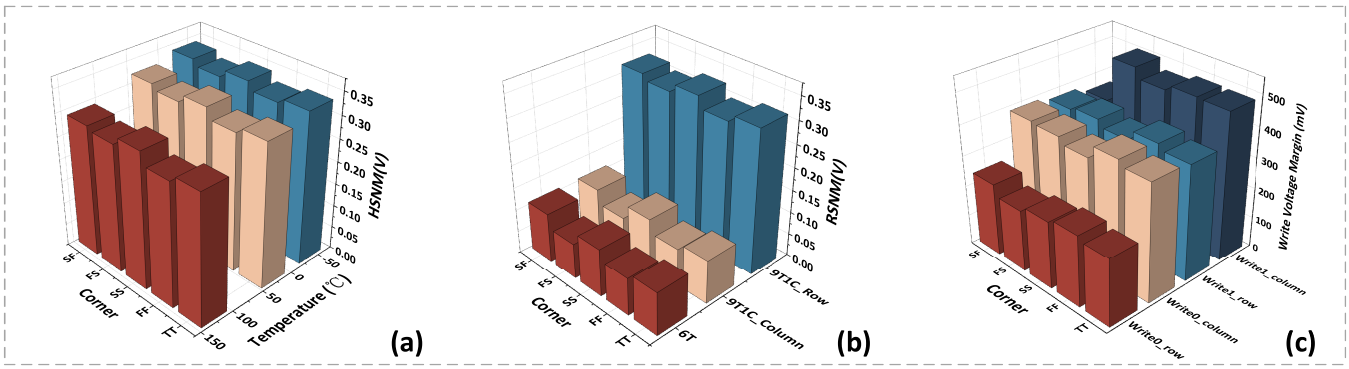


Fig. 8. Stability and write-margin analyses. (a) HSNM of the proposed 9T1C cell under different temperatures and process corners. (b) RSNM comparison of row-wise read and column-wise read in the proposed 9T1C cell with conventional 6T read under different process corners. (c) Bidirectional row-wise and column-wise write voltage margins of the proposed 9T1C cell.

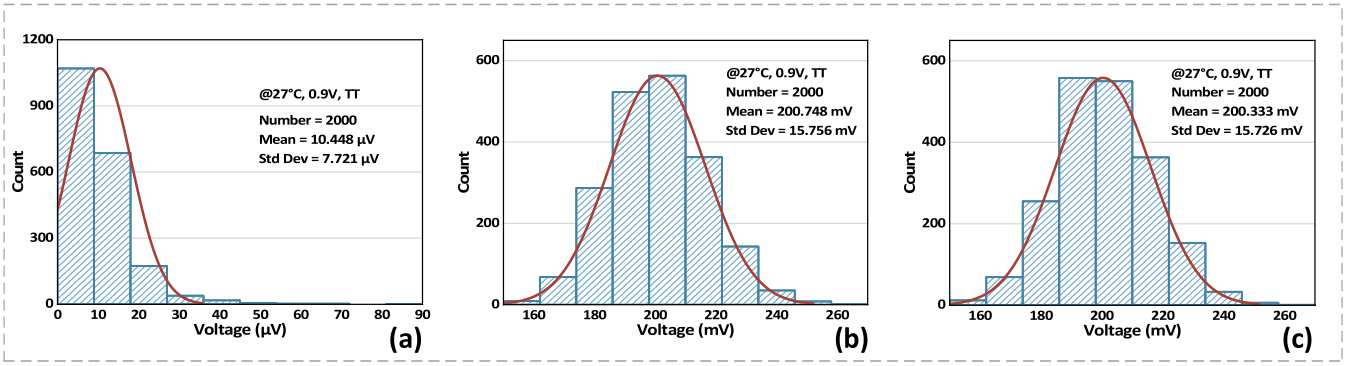


Fig. 9. Monte Carlo read-disturb analysis. (a) Row-wise reading "0" in the proposed 9T1C cell. (b) Column-wise reading "0" in the proposed 9T1C cell. (c) Conventional 6 T reading "0".

bit remains within a narrow range, varying from 2.87 to 3.26 fJ/bit. This result indicates that the proposed MAC operation maintains stable energy efficiency under process variation.

Fig. 11(b) shows the energy efficiency under different supply voltages at the TT corner and 27 °C. As VDD decreases from 0.9 to 0.7 V, the overall energy per bit decreases from 2.87 to 1.51 fJ/bit. This trend confirms stable and energy-efficient operation across a wide supply-voltage range

$$|INL(M)| = \left| \frac{V_{ideal}(M) - V_{out}(M)}{LSB} \right|. \quad (2)$$

The output linearity is shown in Fig. 11(c). The absolute INL is defined in (2), where $V_{out}(M)$ is the simulated output voltage at the MAC value M, and $V_{ideal}(M)$ is the theoretically expected output voltage used as the ideal reference line. The LSB is defined according to the voltage interval between two adjacent ideal MAC output levels. As the MAC value increases from 0 to 960, the output voltage increases correspondingly, showing a nearly linear transfer characteristic over the full output range. The calculated absolute INL remains within 0.026–0.08 LSB, indicating that the deviation from the ideal linear response is very small. This result confirms good output linearity of the proposed MAC operation.

To further verify the robustness of the MAC operation, 2000 Monte Carlo simulations were carried out for three representative outputs, that is, 0, 480, and 960, as shown in Fig. 11(d)–(f). For output 0, the mean output voltage is −18.05 mV with a standard deviation of 169.07 μV. For output 480, the mean output voltage is 438.56 mV with a standard deviation of 336.55 μV. For output 960, the mean output voltage is 895.16 mV with a standard deviation of 90.98 μV. In all three cases, the output distributions remain well concentrated around their expected values, confirming good robustness under process variation. In addition, the scalability of the proposed cell was further evaluated under larger array configurations, which supports its potential applicability to larger capacity designs.

To further benchmark the proposed design, Tables I and II compare its Boolean logic and MAC performance with prior SRAM-CIM macros.

Table I compares the proposed design with prior Boolean logic SRAM-CIM macros. The proposed 9T1C macro achieves 1.218 fJ/bit logic energy at 0.9 V, which is lower than all the compared works. Meanwhile, it reaches the highest operating frequency of 1000 MHz and the highest throughput of 1024 GOPS/Kb among the listed designs. The corresponding energy efficiency is 821 TOPS/W. It should be noted that the reported logic energy of the proposed macro corresponds to the directly supported AND, OR, and IMP operations. In addition, the proposed macro maintains self-cycling capability while achieving lower energy and higher speed, demonstrating its advantage for high-throughput Boolean logic computing.

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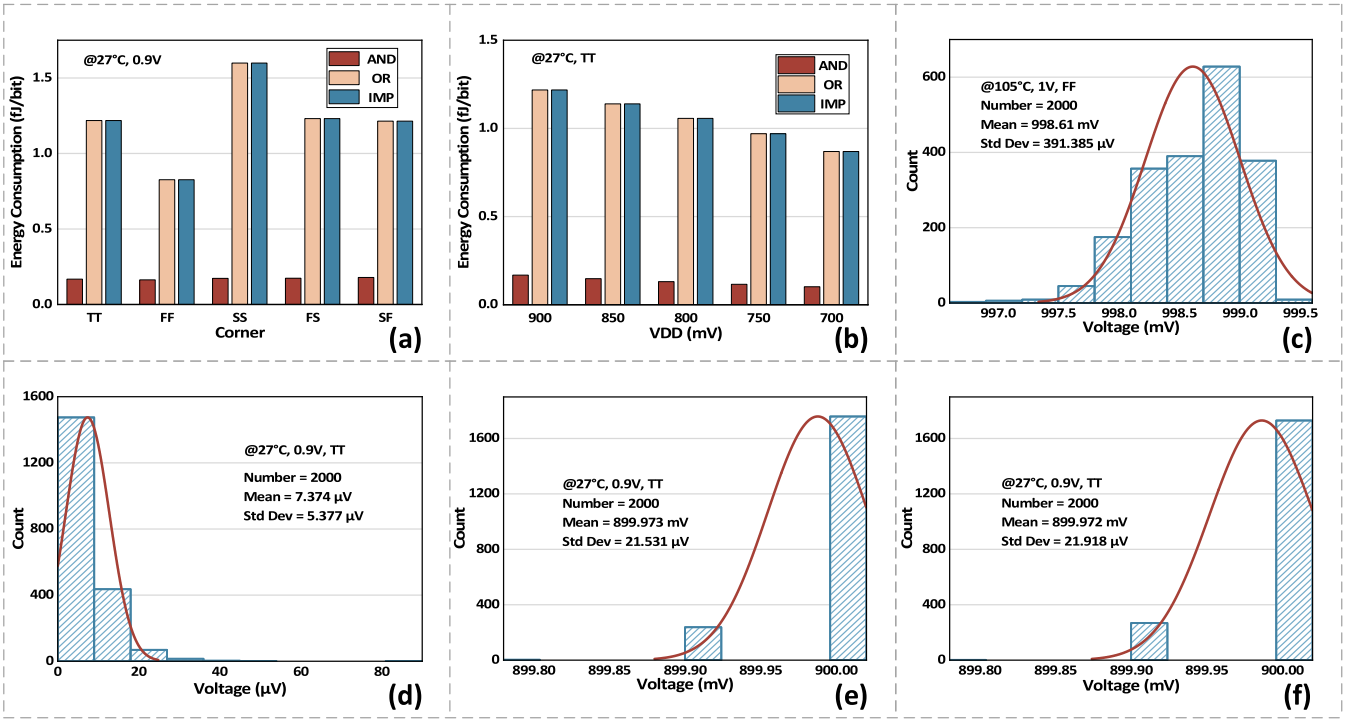


Fig. 10. Energy consumption, dynamic-node retention, and Monte Carlo results of the self-cycling Boolean logic operations. (a) Energy consumption of AND, OR, and IMP under different process corners. (b) Energy consumption of AND, OR, and IMP under different VDDs. (c) Cell storage state after a 1-ns nonstable interval with Q = 1 under the worst-case leakage condition. (d) 2000 Monte Carlo distribution of the output voltage for 0 AND 1. (e) 2000 Monte Carlo distribution of the output voltage for 1 OR 0. (f) 2000 Monte Carlo distribution of the output voltage for 0 IMP 1.

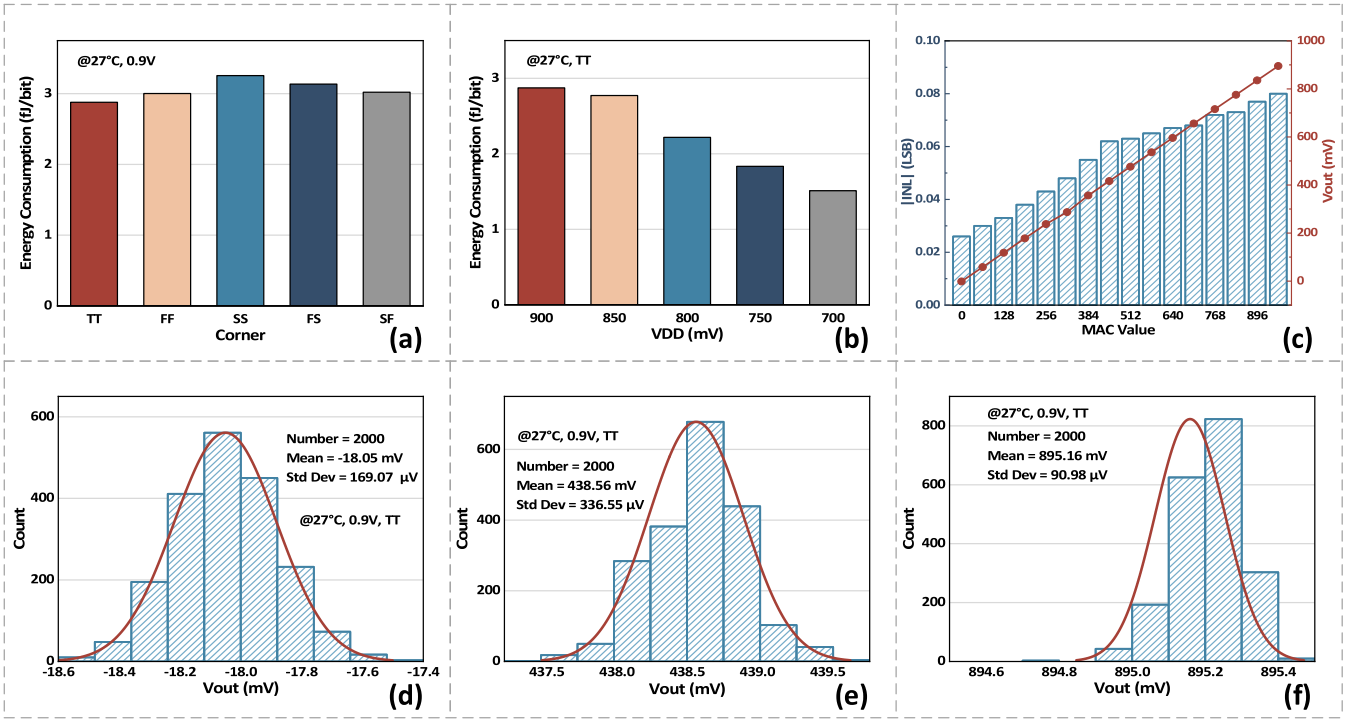


Fig. 11. Energy consumption, linearity, and Monte Carlo results of the MAC operation. (a) Energy consumption under different process corners. (b) Energy consumption under different VDDs. (c) Output voltage and INL versus MAC value. (d) 2000 Monte Carlo distribution at output 0. (e) 2000 Monte Carlo distribution at output 480. (f) 2000 Monte Carlo distribution at output 960.

Table II summarizes the comparison with prior MAC-oriented SRAM-CIM macros. For the native 4-bit-input/1-bit-weight MAC cycle, the proposed design achieves a high throughput of 1170.3 GOPS, mainly benefiting from its 64 output channels. In addition, the proposed 9T1C cell maintains a compact bit-cell area

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TABLE I
PERFORMANCE COMPARISON WITH PRIOR BOOLEAN LOGIC SRAM-CIM MACROS

Boolean Logic	*****This Work***	****JSSC'22[18]***	*****VLSI'25[19]***	*****TCAS1'24[24]***	*****VLSI'22[25]***	****ISSCC'22[26]***
Technology	28nm	28nm	55nm	28nm	28nm	28nm
Macro Size	4Kb	16Kb	4Kb	16Kb	16Kb	32Kb
Cell Type	9T1C	8T	10T	10T1C	6T/8T	6T
Supply Voltage (V)	0.7-0.9	0.65	1.2	0.9	0.5-0.9	0.8
Frequency (MHz)	1000	113 (0.75V)	823.7	385	600 (0.6V)	333
Logic Energy (fJ/bit)	1.218 (0.9V)	3.7 (0.66V)	4.1	14.03	7.8 (0.6V)	19.21-35.55
Energy Efficiency (TOPS/W)	821 (0.9V)	270.5 (0.66V)	246.9	71.28	128.2 (0.6V)	27.38
Throughput (GOPS/Kb)	1024	115.7 (0.75V)	843	394.2	39	11.8
Self-cycling	YES	YES	YES	YES	NO	NO

**** Measured silicon results. ** Simulation results.***

TABLE II
PERFORMANCE COMPARISON WITH PRIOR MAC-ORIENTED SRAM-CIM MACROS

MAC	*****This Work***	*****TCAS1'24[24]***	****TCAS2'23[27]***	****JSSC'20[28]***	****ISSCC'21[29]***	*****TCAS2'24[30]***
Technology	28nm	28nm	28nm	7nm	28nm	28nm
Macro Size	4Kb	16Kb	32Kb	4Kb	384Kb	4Kb
Cell Type	9T1C	10T1C	9T1C	8T	6T+LCC	9T1C
Bit-cell area (μm^2)	0.91	2.85	0.994	0.053	0.245[1]	N/A
Input/Weight/Output	4/1/4	2/2/(N/A)	4/4/4	4/4/4	4/4/12	4/4/4
Output Channels	64	N/A	64	16	16	16
Frequency (MHz)	142.9	196	100	183.5	250	222
Throughput (GOPS)	1170.3	802.8	1638.4	372.4	128	455
Energy Efficiency (TOPS/W)	696-1323.5[2]	6.6	646.6[2]	351[2]	60.3-94.3[2]	1012[2]

[1]Average area considered local computing cell . [2]Energy efficiency is calculated by counting one MAC operation as two operations.* Measured silicon results. ** Simulation results.

of 0.91 μm^2 in 28-nm CMOS. Its energy efficiency reaches 696–1323.5 TOPS/W, which is competitive with prior designs. These results indicate that the proposed macro provides a good balance among throughput, parallelism, area efficiency, and energy efficiency for MAC computing.

In conclusion, compared with [18], the proposed 9T1C macro preserves bidirectional access and self-cycling Boolean logic in a smaller cell topology. Compared with [24], the proposed design maintains charge-domain MAC capability with a reduced cell size, while further integrating bidirectional read/write and self-cycling Boolean logic operations, including AND, OR, and IMP. Moreover, the proposed macro achieves competitive or improved performance in both Boolean logic and MAC-oriented comparisons, as summarized in Tables I and II.

IV. CONCLUSION

This article presented a 28-nm 4-Kb 9T1C SRAM-CIM macro that unifies bidirectional read/write, full-array self-cycling Boolean logic operations, and MAC operation within a compact bit-cell structure. The proposed design achieves improved read stability through the row-wise read path, low-energy and robust Boolean logic operations, and charge-domain MAC operation with good output linearity. Simulation results show logic energy as low as 1.218 fJ/bit, Boolean logic throughput of 1024 GOPS/Kb, and MAC throughput of 1170.3 GOPS. Therefore, the proposed macro provides a

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compact and efficient multifunctional SRAM-CIM solution for integrating logic-centric and MAC-oriented computing in a single architecture.

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